The Impact of TOU on PV—Go West

by John Supp and
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This 3.6 kW panel was installed under the "Rebuild a Greener San Diego" program for those who lost their homes in the 2003 fires.

Today, most residences are billed for electricity usage according to the total amount used over a given month. In the San Diego area, most residential customers are charged for their electricity use under the standard domestic residential, or DR, tariff applied by the local utility, San Diego Gas and Electric Company (SDG&E). Under this tariff, customers pay a set amount for the baseline usage—about $0.13/kWh with a baseline quantity of usually 300-500 kWh—and an increasing amount for each block, or tier, of electricity used thereafter. The subsequent tiers include up to 130% of baseline, up to 200% of baseline, 200%-300% of baseline, and over 300% of baseline. For this last tier, customers pay about $0.27/kWh. It does not matter at what time of day the home uses that electricity; all that matters is the total amount used by the month's end.

As of January 1, 2007, PV homeowners who sign up for the California Solar Initiative (CSI) are required to switch to a domestic time-of-use (TOU) rate. In addition to the traditional seasonal variations in the price of electricity, under this rate structure the electricity bill will be calculated based on whether a customer buys power on peak (between 12 pm and 6 pm) or off peak (between 6:01 pm and 11:59 am). What has always mattered to electric generation and distribution companies—the time of day that power gets used and generated—will now matter to the customer, too.

We modeled the effect of PV on the bill under these two rate structures for a typical residential electricity customer, using data from the software program, PV Watts 2, and Itron's eShape, a database of load profiles for residential electricity and gas usage. We also determined the size of the PV system that would be needed to approach zeroing out the monthly electric bill for a typical residential customer.

For our modeling exercise, we used the most common profile for a San Diego region home from Itron's eShape—a residential, southwest-facing, single-family home with central air conditioning and gas heat. This profile uses an inland designation for purposes of establishing baseline electricity allotment, which assumes some on-peak use of air conditioning during the summer. (The translation from Itron's usage curve into data points introduces a potential small margin of error.) System design assumptions include a 15° tilt, true azimuth readings, a 5.5 kW DC array, 92020 zip code and a 0.75 derate factor for PV Watts 2.

Under the DR tariff, a 5.5kW DC system was needed to generate enough electricity so that the customer would end up owing almost nothing for electricity usage (see Table 1). With the new SDG&E TOU rate structure, we found that the same-sized PV system yielded a different financial outcome (see Table 2). In the first scenario, the south-facing array outperformed the west-facing one. In the second scenario with the DR-TOU tariff, the scenarios are reversed. The west-facing array outperforms the south-facing array.

Additional savings would result from shifting loads from on peak times to off peak times. This does not mean using any less electricity—although that would be useful as well—but rather changing the time that common high usage items are used. Using the same PV setup and azimuths, but reducing the on-peak usage to 20% during summer, spring, and autumn, allows a customer to save an additional $90 per year for the south-facing setup, and an additional $24 per year for the west-facing one (see Table 3). Because south- and west-facing systems have a tilt and azimuth that maximizes summertime performance, and because they produce most of their power during the 12 pm-6 pm

Table 1. PV Effect Under DR Tariff

			West-facing (270°)		South-facing (180°)	
	kWh used per mo.	DR no PV	PV kWh	Net Bill	PV kWh	Net Bill
Summer	925	$ 177	799	$ 16	789	$ 17
Autumn	740	$ 130	635	$ 14	681	$ 8
Winter	550	$ 83	356	$ 25	481	$ 9
Spring	705	$ 119	561	$ 19	626	$ 10
Yearly Total		$1,527		$222		$132

Table 2. PV Effect Under TOU Tariff

			West-facing (270°)		South-facing (180°)	
	kWh used per mo.	On Peak usage %	PV kWh	Net Bill	PV kWh	Net Bill
Summer	925	41	799	$ 4*	789	$ 22
Autumn	740	33	635	$ 0*	681	$ 9
Winter	550	20	356	$ 25	481	$ 9
Spring	705	33	561	$ 14	626	$ 14
Yearly Total				$129		$162

* A minimum charge from SDG&E of approximately $0.17/day may apply.

Table 3. Effect of Shifting Peak Load

			West-facing (270°)		South-facing (180°)	
	kWh used per mo.	On Peak usage %	PV kWh	Net Bill	PV kWh	Net Bill
Summer	925	20	799	$ (8)*	789	$ 8
Autumn	740	20	635	$ (6)*	681	$ 0*
Winter	550	20	356	$ 25	481	$ 9
Spring	705	20	561	$ 10	626	$ 7
Yearly Total				$105		$72

* A minimum charge from SDG&E of approximately $0.17/day may apply.

peak window, a customer may save more under the TOU tariff than under the DR tariff.

By shifting some energy uses to off-peak times, a well-sited, appropriately sized PV system could end up costing less in 2007 than it would have in 2006—even though per-watt rebate levels have dropped and PV costs continue to fluctuate.

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